Experiment 9:

Aim: : To demonstrate the performance of Naïve Bayes algorithm by using iris dataset

Program:

```
# Import libraries
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns

# Load Iris dataset
iris = load_iris()

# Features and Target
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Naïve Bayes model
model = GaussianNB()

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:\n")
print(cm)

# Classification Report
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

# Plot Confusion Matrix
plt.figure(figsize=(6,5))
sns.heatmap(cm,
            annot=True,
```

```
        fmt='d',
        cmap='Blues',
        xticklabels=iris.target_names,
        yticklabels=iris.target_names)

plt.title("Naïve Bayes - Iris Dataset")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

```
Accuracy: 0.9777777777777777

Confusion Matrix:

[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

Classification Report:

              precision    recall  f1-score   support

           0       1.00      1.00      1.00        19
           1       1.00      0.92      0.96        13
           2       0.93      1.00      0.96        13

    accuracy                           0.98        45
   macro avg       0.98      0.97      0.97        45
weighted avg       0.98      0.98      0.98        45
```